# WASP-2b

**R-0155**

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-2_250613` · created 2026-07-16T14:37:02+00:00

**BENCHMARK: ACCURATE — fit consistent with published values**

## Results

| | |
|---|---|
| **Mid-Transit Time (Tmid)** | 2460839.9006 +/- 0.0082 BJD_TDB |
| **Ratio of Planet to Stellar Radius (Rp/R*)** | 0.156 +/- 0.042 |
| **Transit depth (Rp/Rs)^2** | 2.42 +/- 1.31 [%] |
| **Orbital Inclination (inc)** | 82.98 +/- 2.82 |
| **Airmass coefficient 1 (a1)** | 0.936 +/- 0.014 |
| **Airmass coefficient 2 (a2)** | 0.054 +/- 0.018 |
| **Scatter in the residuals of the lightcurve fit is** | 1.54 % |
| **Best Comparison Star** | #2 - [71, 437] |
| **Optimal Method** | PSF photometry |
| **Transit Duration (day)** | 0.045 +/- 0.037 |

## Accuracy vs published ephemeris (ExoClock/NEA)

| | |
|---|---|
| **Rp/R■ fit vs published** | 0.156 ± 0.042 vs 0.1326 (deviation 0.52σ) |
| **Duration fit vs published** | 0.045 d vs 0.0737 d (deviation 0.72σ) |
| **Mid-time O–C** | 21.5 min |
| **Depth SNR** | 3.4 |
| **Residual scatter** | 1.54 % |

## Light curve

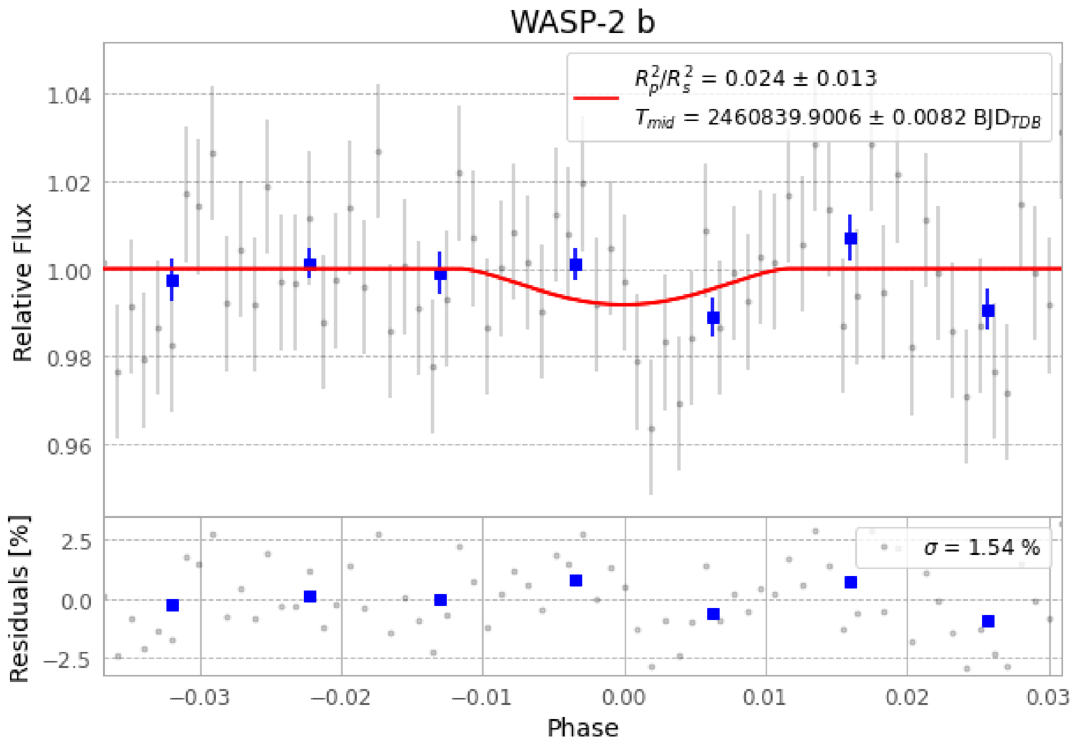


Figure 1 — FinalLightCurve_WASP-2 b_2025-06-13.png (SHA-256 3c6b48c94880969c…)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [327, 206], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 293a507cc6e2783f…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 6eeacf9`

## Data provenance

71 FITS frames (47 MB), `WASP-2250613073615.FITS` → `WASP-2250613110615.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-250613.log (9bbcfbc13ded…), FinalLightCurve_WASP-2 b_2025-06-13.png (3c6b48c94880…), FinalLightCurve_WASP-2 b_2025-06-13.csv (735e9f10425e…)

## Compute resources

Wall time 150.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)